CS 610 Semester 2024–2025-I: Assignment 3

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NOTE:

All codes attached along with this assignment have been executed on KD first floor lab workstation with ID: **CSEWS179**

Problem 1

The time taken by the various implementations of the matrix multplication problem are summarised below. Time values are in ms.

Table 1: Summary of time taken for different implementations of Matrix Multiplication

SNo.	**N**	**Sequential**	**SSE4 (Un-aligned)**	**SSE4 (Aligned)**	**AVX2 (Un-aligned)**	**AVX2 (Aligned)**	**AVX2 (V1)**	**AVX2 (V2)**
1	256	18	4	4	1	1	0	0
2	512	126	30	30	15	15	4	3
3	1024	1276	316	317	159	158	49	31
4	2048	9324	2614	2593	1804	1721	395	252
5	4096	11073	2604	2741	1678	1772	393	250

- **AVX2 (V1)** is AVX2 implementation along with blocking of 512×16 (Chosen through experimentation with different block sizes).
- **AVX2 (V2)** is AVX2 implementation along with blocking of 512×24 (Chosen through experimentation with different block sizes).

An unaligned AVX2 implementation with 512×24 blocking gives the best performance across different values of N.
Compilation Command: `g++ -g -std=c++17 -masm=att -march=native -O2 -fopenmp -fverbose-asm -fno-asynchronous-unwind-tables -fno-exceptions -fno-rtti problem1.cpp -o problem1.out`

Problem 2

The time taken by the various implementations of the prefix problem are summarised below. Time values are in ms.

Table 2: Summary of time taken for different implementations of Inclusive Prefix Sum

SNo.	**N**	**Serial**	**OMP**	**SSE**	**AVX2**
1	2^{16}	60	40	54	39
2	2^{30}	495940	473691	506676	460777

Compilation Command: `g++ -g -std=c++17 -masm=att -march=native -O2 -fopenmp -fverbose-asm -fno-asynchronous-unwind-tables -fno-exceptions -fno-rtti problem2.cpp -o problem2.out`

Problem 3

Explanations are given as comments alongside the code. Test cases (in `./testcase` directory) and a log file for one of the test cases is attached.
Compilation Command: `g++ -g -std=c++17 -masm=att -march=native -O2 -fopenmp -fverbose-asm -fno-asynchronous-unwind-tables -fno-exceptions -fno-rtti problem3.cpp -o problem3.out`
Run Command: `./problem3.out file2.txt 15 10 14 output.txt`

Problem 4

Part (i)

The following optimizations were made:

- **Common Subexpression Elimination**
  There are multiple instances where the same operations are being done many times. Instead of using `xi` values, the actual values are substituted. This reduces repeated arithmetic computations over a long loop nest thereby increasing speed up.
  Similarly, multipliers that have to be computed again are also stored in an array before the loop begins.

- **Removing long conditionals**
  The big conditional statement where `q1` to `q10` are compared with `e1` to `e10` is removed. Instead negation of each specific condition is checked immediately after the corresponding `q` is calculated. If the condition is not met the entire iteration is skipped using the `continue` keyword.

Speed Up:
Time taken by code without optimizations = 298.680 seconds
Time taken by code with optimizations = 144.947 seconds
Speed Percentage = $\left(\frac{298.680-144.947}{298.680}\right) \times 100 =$ **51.47%**
Compilation Command: `gcc -O3 -std=c17 -D_POSIX_C_SOURCE=199309L problem4-v0-p1.c -o prob4-v0.out`

Part (ii)

Applied the `reduction` and `collapse` directives on the outermost loop on top of the optimizations done in Part (i). Time taken by code without optimizations = 298.680 seconds
Time taken by code with optimizations and OpenMP = 40.255 seconds
Speed Ratio = $\left(\frac{298.680}{40.255}\right) =$ **7.42**
Compilation Command: `gcc -O3 -std=c17 -D_POSIX_C_SOURCE=199309L problem4-v0-p2.c -o prob4-v0.out -fopenmp`